Supplementary Information

1. Supplementary Table S1: List of human hormones[1,2].

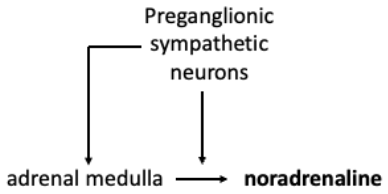
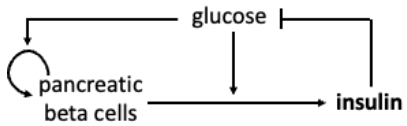
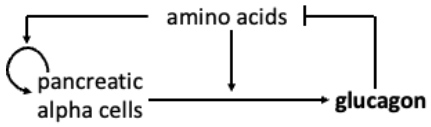
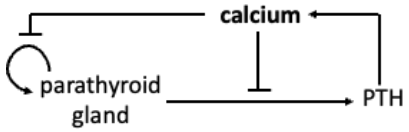
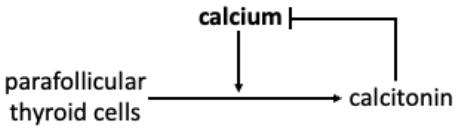
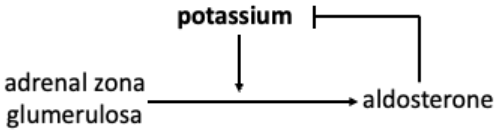
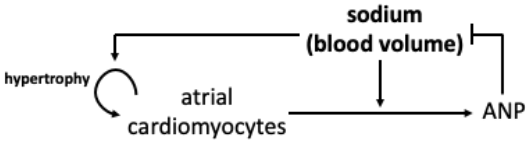
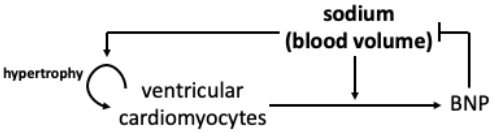
List of all human hormones	
1. adiponectin	not included in Table S2
2. adrenaline	included in Table S2
3. adrenocorticotropic hormone (ACTH)	included in Table S2
4. aldosterone	included in Table S2
5. alpha-melanocyte stimulating hormone (alpha-MSH)	not included in Table S2
6. amylin	not included in Table S2
7. androgens	included in Table S2
8. angiotensin	not included in Table S2
9. antidiuretic hormone (ADH)	included in Table S2
10. apelin	not included in Table S2
11. atrial natriuretic peptide (ANP)	included in Table S2
12. beta- endorphins	included in Table S2
13. brain natriuretic peptide (BNP)	included in Table S2
14. c-type natriuretic peptide (CNP)	not included in Table S2
15. calcitonin	included in Table S2
16. calcitonin-gene related peptide	not included in Table S2
17. cholecystokinin (CCK)	included in Table S2
18. corticotropin-releasing hormone (CRH)	included in Table S2
19. cortisol	included in Table S2
20. endothelin	included in Table S2
21. erythropoietin	included in Table S2
22. estrogens	included in Table S2 (listed as estradiol)
23. follicle-stimulating hormone (FSH)	included in Table S2
24. galanin	not included in Table S2
25. gastric inhibitory polypeptide (GIP)	included in Table S2
26. gastrin	included in Table S2
27. ghrelin	included in Table S2
28. glucagon	included in Table S2
29. glucagon-like peptide 1 (GLP-1)	not included in Table S2

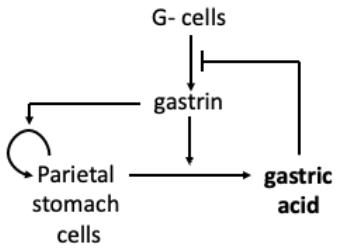
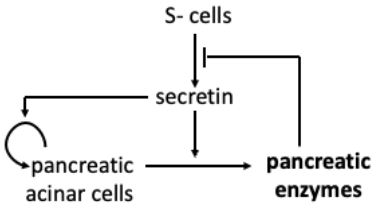
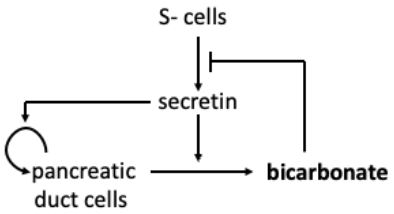
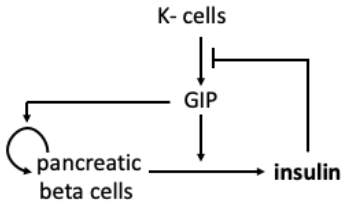
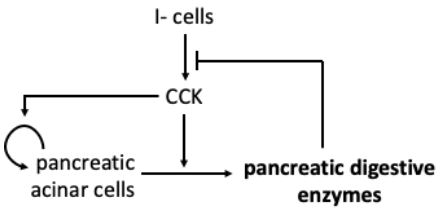
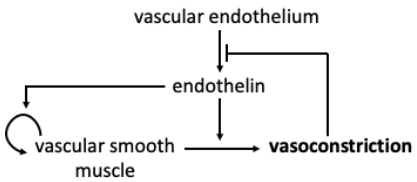
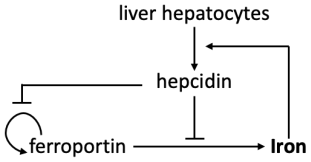
30. glucagon-like peptide 2 (GLP-2)	not included in Table S2
31. gonadotropin releasing hormone (GnRH)	included in Table S2
32. growth hormone (GH)	included in Table S2
33. growth hormone releasing hormone (GHRH)	included in Table S2
34. hepcidin	included in Table S2
35. insulin	included in Table S2
36. insulin-like growth factor 1 (IGF-1)	included in Table S2
37. leptin	not included in Table S2
38. luteinizing hormone (LH)	included in Table S2
39. melanocyte-stimulating hormone (MSH)	included in Table S2
40. melatonin	included in Table S2
41. motilin	not included in Table S2
42. neurotensin	included in Table S2
43. noradrenaline	included in Table S2
44. osteocalcin	not included in Table S2
45. oxytocin	included in Table S2
46. parathyroid hormone (PTH)	included in Table S2
47. parathyroid-hormone related protein (PTHrP)	not included in Table S2
48. pancreatic polypeptide (PP)	not included in Table S2
49. peptide histidine-isoleucine (PHI)	not included in Table S2
50. peptide YY (PYY)	not included in Table S2
51. pituitary adenylate cyclase activating polypeptide (PACAP)	not included in Table S2
52. progesterone	included in Table S2
53. prolactin	included in Table S2
54. renin	not included in Table S2
55. secretin	included in Table S2
56. somatostatin	included in Table S2
57. testosterone	included in Table S2
58. T4	included in Table S2
59. TSH	included in Table S2
60. thrombopoietin	included in Table S2
61. TRH	included in Table S2

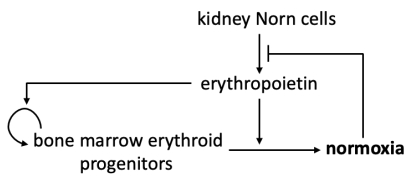
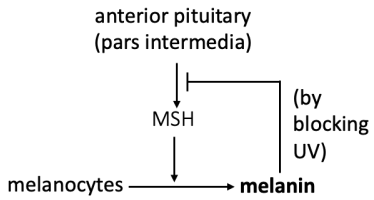
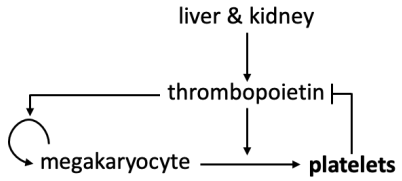
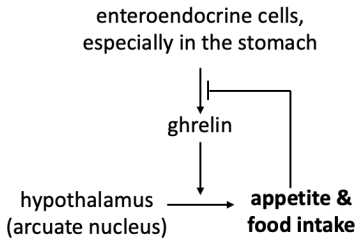
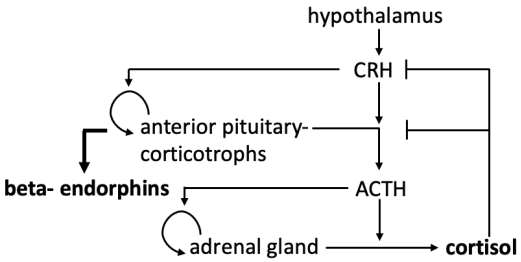
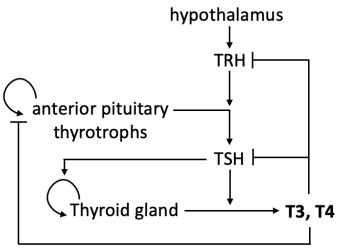
62. T3	included in Table S2
63. vasoactive intestinal polypeptide (VIP)	not included in Table S2

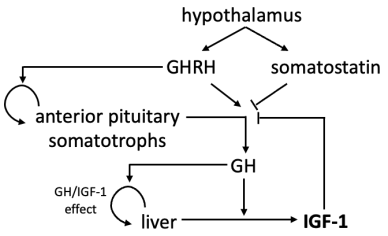
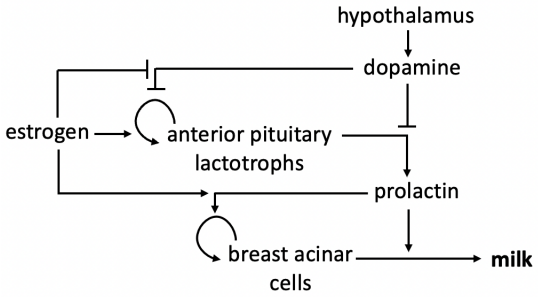
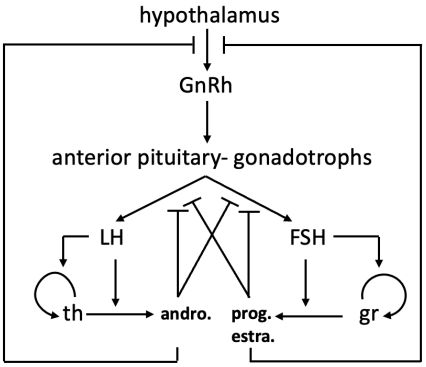
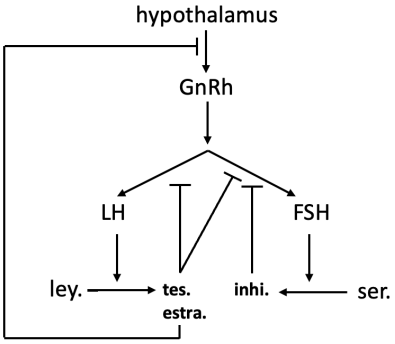
2. Supplementary Table S2: Examples for each endocrine circuit class. The main physiological output of each circuit is in bold.

Hormone/ metabolite	Class	Circuit design	References and notes
Antidiuretic hormone (ADH)	1	posterior pituitary → **anti-diuretic hormone (ADH)**	3
Oxytocin	1	posterior pituitary → **oxytocin**	4
Melatonin	1	pineal gland → **melatonin**	5
Neurotensin	1	enteroendocrine cells, predominantly in the small intestine → **neurotensin**	6
Adrenaline	2	Preganglionic sympathetic neurons → adrenal medulla → **adrenaline**; Preganglionic sympathetic neurons → adrenal medulla	7

Noradrenaline	2		8
Insulin	3		9
Glucagon	3		10–12
Parathyroid hormone (PTH)	3		9,13
Calcitonin	3		14 This system seems to be missing the gland mass regulation of other class 3 circuits.
Aldosterone	3		15 This system seems to be missing the gland mass regulation of other class 3 circuits.
Atrial natriuretic peptide (ANP)	3		16
Brain natriuretic peptide (BNP)	3		17,18

Gastrin	4		19
Secretin	4		20,21
Secretin	4		20,21
Gastric inhibitory polypeptide (GIP)	4		22,23
Cholecystokinin (CCK)	4		24–26
Endothelin	4		27,28
Hepcidin	4		29

Erythropoietin	4		30
Melanocyte-stimulating hormone (MSH)	4		31,32 This system seems to be missing the gland mass regulation of other class 4 circuits.
Thrombopoietin	4		33–35
Ghrelin	4		36 This system seems to be missing the gland mass regulation of other class 4 circuits.
Corticotropin-releasing hormone (CRH); Adrenocorticotropic hormone (ACTH); Beta- endorphins; Cortisol	5a		37
Thyrotropin-releasing hormone (TRH); Thyroid-stimulating hormone (TSH); Triiodothyronine (T3), Tetraiodothyronine (T4)	5b		38

Growth hormone-releasing hormone (GHRH); Somatostatin; Growth hormone (GH); Insulin-like growth factor 1 (IGF-1)	5c		39–48
Prolactin	5d		49,50
Gonadotropin-releasing hormone (GnRH); Luteinizing hormone (LH); Follicle-stimulating hormone (FSH); Androgens; Progesterone; Estradiol-female	5e	 Abbreviations stand for (from left to right)- theca cells, androgens, progesterone, estradiol, granulosa cells.	51
Gonadotropin-releasing hormone (GnRH); Luteinizing hormone (LH); Follicle-stimulating hormone (FSH); Testosterone; Estradiol; Inhibin-male	5e	 Abbreviations stand for (from left to right): Leydig cells, testosterone, estradiol, inhibin, sertoli cells	51

3.

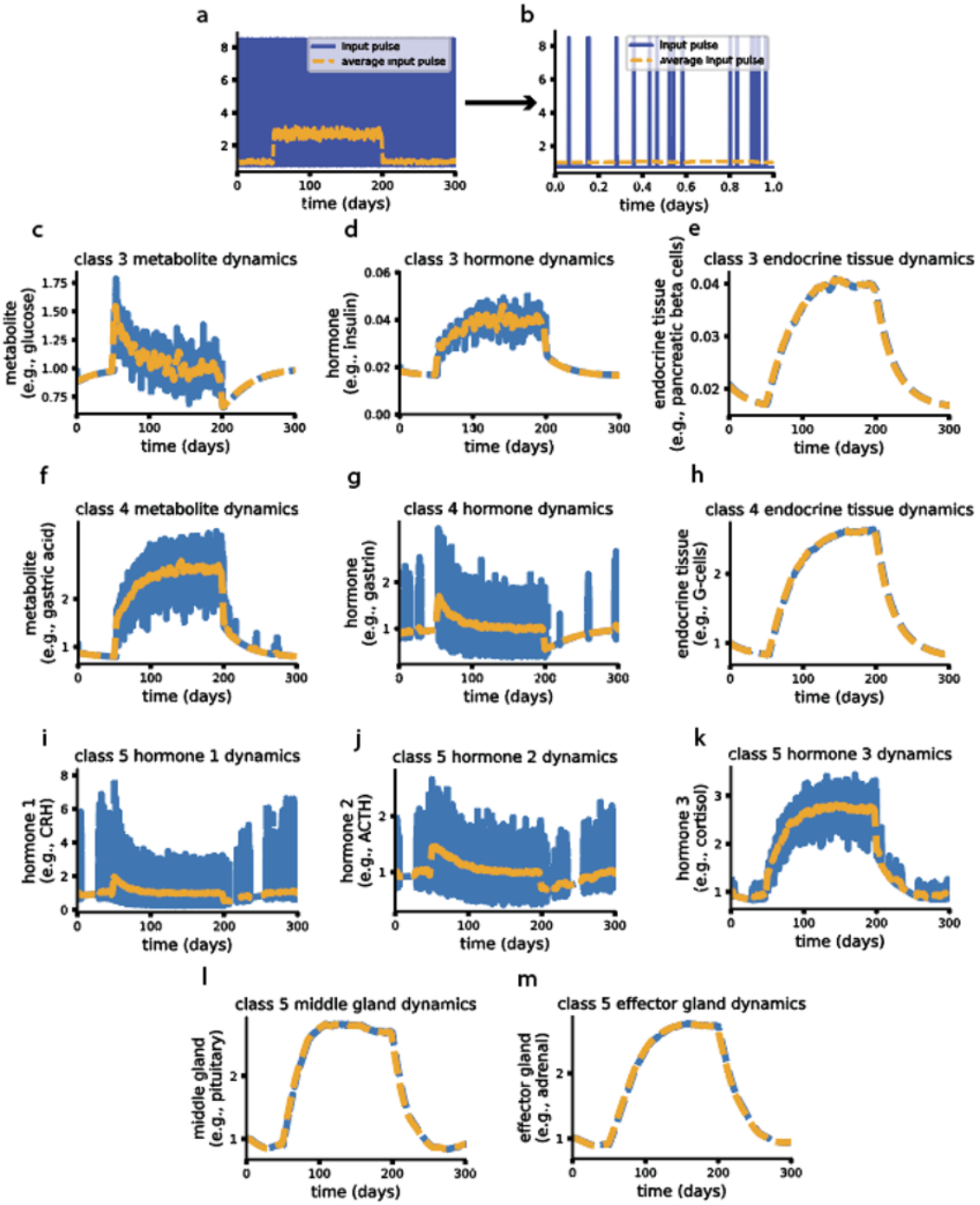


Figure S1: Class 3-5 responses to train input pulses that change rapidly on the scale of minutes. a. Input pulse trains (blue) consist of individual pulses of about 3 minutes each, and occur stochastically with an average frequency that changes on a slow time scale (weeks). The orange curve shows a 1-day moving average of the pulses. b. Zoom in on panel a. (c-m) Responses of the circuits for (c-e) class 3, (f-h) class 4 and (i-m) class 5. Orange curves represent the 1-day moving average as in panel a. Class dynamics qualitatively match those of square input pulses as can be seen in the main text (Figure 1).

4.

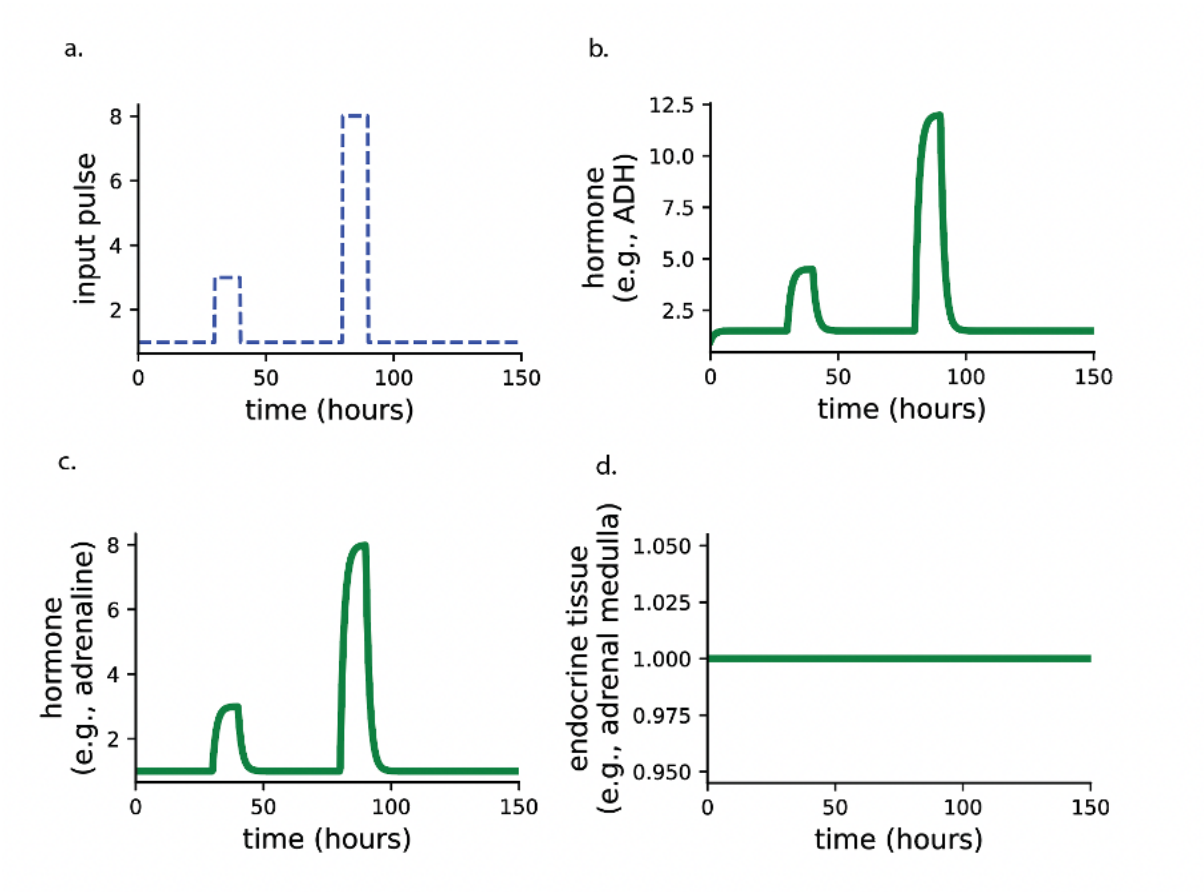


Figure S2: Short-timescale (minutes-hours) simulations of class 1 and 2 circuits show a direct input-output response. a. the input pulse used in the simulations: b. class 1 circuit. c-d. class 2 circuit.

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